Data Analytics Capstone

Technical Declines in India's Unified Payments Interface: Effects of Volume Shifts and Regulatory Interventions

Synopsis

Karthikeyan S

Walsh College

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Sharat Srivatsa

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Introduction

Background and Context

The Unified Payments Interface (UPI) is the real-time retail payment system operated by the National Payments Corporation of India (NPCI). It lets customers of different banks move money instantly through third-party apps, and it has become the dominant way Indians pay for everyday goods and services. Academic work on UPI has largely explained how it grew and why people adopted it (Balakrishnan, 2023; Gopika et al., 2026). Growth has been steep. Using NPCI's monthly member tables (NPCI, n.d.), the attempted volume of the 50 largest remitter members rose from 12.56 billion transactions in January 2024 to 26.75 billion in August 2026, so the system now carries more than twice the load it did at the start of the study window.

Every UPI transaction that fails is classified by NPCI as a business decline (BD) or a technical decline (TD). A BD reflects a customer-side cause such as a wrong PIN or insufficient balance. A TD reflects a failure in the system itself, such as a connectivity, switch, or bank-side processing problem. The two are reported separately for each member bank, together with the approved share, so that approved, BD, and TD percentages sum to 100. TD is the more consequential measure for operators, because it is the part of failure that the operator, rather than the customer, can reduce. Payment-system regulators elsewhere treat availability as a core service-quality target; the Reserve Bank of Australia, for example, reports retail payment services with average availability of at least 99.8 per cent each quarter (Griffiths & Joyce, 2024).

Two forces make TD a live operational question in India. The first is load: when volume rises abruptly, a bank's processing capacity may be strained. The second is regulation. On April

12, 2025, a UPI outage was attributed by NPCI to an unexpected surge in Check Transaction Status application programming interface (API) calls (Tyagi, 2025). NPCI then announced measures on May 21, 2025, effective August 1, 2025, that cap balance checks at 50 per user per day per app, limit linked-account views to 25 per customer per day, allow at most three transaction-status checks in two hours with a minimum 90-second gap, and confine AutoPay mandate execution to non-peak hours (Khanduri & Mittal, 2025). The stakeholders are NPCI, which sets the rules; remitter banks, which provision capacity; and the third-party apps whose API behavior generates much of the traffic.

Literature Review and Research Gap

Research on UPI sits in three streams. The first explains growth and adoption. Balakrishnan (2023) traces the expansion of digital payments in India since 2010, identifies UPI as the main driver, and compares it with Brazil's PIX to draw lessons for other countries. Gopika et al. (2026) examine the factors associated with UPI growth in a time-series framework and link them to government-led design of the ecosystem. These studies treat volume as the outcome. In this study, volume is the exposure, and service quality is the outcome.

The second stream concerns reliability and failure. Chawla et al. (2021) relate banking regulations to online payment failures, which is the closest topic to this study, but it predates the 2024–2026 surge in volume and the 2025 API controls and does not use bank-month panels of technical declines. Central-bank work on payment reliability, such as the Reserve Bank of Australia's account of retail payment availability (Griffiths & Joyce, 2024), shows how operators monitor service quality at an aggregate level, and shows that a few severe incidents can dominate a quarter's record. Commentary on India's April 2025 outage ties it to API call volume (Tyagi,

2025), and legal commentary describes the controls that followed (Khanduri & Mittal, 2025). These sources describe events; none estimates their effect on bank-level technical declines.

The third stream is methodological. Interrupted time series is the standard design for evaluating a policy introduced at a known date (Bernal et al., 2017; Wagner et al., 2002). Fractional-response models handle outcomes that are proportions (Papke & Wooldridge, 1996). Cluster-robust inference with few clusters needs bootstrap corrections (Cameron et al., 2008). Survival methods suit durations such as spike length (Cox, 1972; Kaplan & Meier, 1958). The contribution here is to combine these tools on one public, member-level dataset for a question operators face. Structured searches of Google Scholar on combinations of UPI, technical decline, failure rate, outage persistence, API controls, and seasonal failure spikes returned no study that estimates the effect of volume shifts and the 2025 controls on technical declines by member bank. The gap is therefore worded as "to our knowledge," and the search log is kept in the repository.

Problem Statement

Chawla et al. (2021) studied how banking regulations relate to online payment failures, but, to our knowledge, no published study has used NPCI's member-level monthly performance tables to estimate how sudden volume shifts affect technical decline rates, or has evaluated the August 2025 API controls against a pre-intervention baseline. Without that evidence, capacity is added where it is assumed to be needed and API controls are judged by anecdote. This research aims to estimate how sudden volume shifts and regulatory interventions affect technical decline rates in UPI, in order to help NPCI and banks decide where to add capacity and which API controls to enforce.

Purpose of the Study

The study is primarily descriptive and explanatory: it estimates the size and direction of associations between TD and volume shocks, calendar peaks, and the timing of the 2025 intervention (without claiming proof of cause), and it characterizes how persistent TD spikes are and how much of TD variation is bank-specific. A secondary, predictive objective benchmarks a machine learning (ML) model against transparent statistical baselines for next-month TD and spike detection. The prediction work supports the explanatory work and does not replace it, and none of the predictive results is read as causal.

Scope and Objectives

The analysis covers the remitter side of UPI for the top 50 member entities by volume, monthly, from January 2024 to August 2026 (32 months). Four research questions, each with testable hypotheses, structure the work. All four use the same dataset.

Research Question 1 (RQ1)

Do sudden volume shifts raise a bank's technical decline rate? Let V_{it} be bank i's attempted volume in month t. A sudden shift is measured by month-over-month log growth, $\Delta \ln V_{it}$, and by NewPeak_{it}, an indicator that V_{it} exceeds the bank's own previous maximum. $H0_{1a}$: the coefficient on $\Delta \ln V_{it}$ is zero; $H1_{1a}$: it is not zero. $H0_{1b}$: the NewPeak coefficient is zero; $H1_{1b}$: it is not. The hypotheses are two-sided because higher load could raise TD, but banks that scale ahead of demand could show no effect or even lower TD.

Research Question 2 (RQ2)

Did system-level TD change after NPCI's API controls took effect on August 1, 2025? $H0_2$: the post-intervention level shift in volume-weighted monthly TD is zero; $H1_2$: it is not zero. A placebo hypothesis, $H0_{2p}$, states that BD, which the controls should not target directly, shows

no comparable shift. Because every bank was exposed on the same date and no untreated group exists, RQ2 is a descriptive timing evaluation; its decision rule is fixed in advance (see Solution to RQ1–4).

Research Question 3 (RQ3)

Do predictable calendar peaks, namely March fiscal year-ends and the October–November festive months, raise TD, and do high-volume banks suffer more? $H0_{3a}$: TD in peak months equals TD in other months for the same bank; $H0_{3b}$: the peak effect does not vary with a bank's volume exposure.

Research Question 4 (RQ4)

How persistent are TD spikes, and how much of TD variation is specific to the bank? $H0_{4a}$: the bank-level share of TD variance (the intraclass correlation, ICC) is zero; $H0_{4b}$: month-to-month autocorrelation of TD within a bank is zero; $H0_{4c}$: spike recovery times do not differ across bank groups.

Sample Size Calculation

Sample size was set for each question using power or confidence-interval logic and then reconciled in one rule: the final sample is the maximum of the question-level minimums. Table 1 summarizes the results. Calculations use $\alpha = 0.05$ (two-sided, $z = 1.960$) and power of 0.80 ($z = 0.842$), so the multiplier is $K = 1.960 + 0.842 = 2.802$.

Table 1
Minimum Sample Size by Research Question

RQ	Method	Key parameters	Minimum N
RQ1	Power (regression; Gelman & Hill, 2007)	$\alpha = .05$, power = .80, smallest effect $\beta = 0.60$ logit per unit $\ln V$; $\sigma_e = 1.051$ (residual SD); within-bank SD of $\ln V = 0.126$. $n = (K\sigma_e / (\beta\, SD_x))^2 = (2.802 \times 1.051 / (0.60 \times 0.126))^2$	1,522 bank-months
RQ2	CI width (ITS)	95% CI half-width target 0.25 pp; residual SD 0.135 pp; lag-1 autocorrelation 0.203 (variance inflation 1.508); 19 pre- and 13 post-break months	26 months
RQ3	Power (Fisher z)	$\alpha = .05$, power = .80, smallest correlation $r = 0.20$: $n = (K / (0.5 \ln((1+r)/(1-r))))^2 + 3$	194 peak bank-months
RQ4	Events per variable	Cox model with 3 covariates × 10 events per variable	30 spike episodes

Note. pp = percentage points; ITS = interrupted time series. Available data: 1,555 bank-months, 32 months, 342 peak bank-months, 69 spike episodes.

As a cross-check, Green's (1991) rule of thumb for a regression with 4 substantive predictors and 86 fixed-effect terms asks for $50 + 8 \times 90 = 770$ observations, which the RQ1 power requirement already exceeds. The binding requirement is RQ1, at 1,522 bank-months, so the final sample size is 1,522. The panel provides 1,555 bank-months after non-bank entities are removed, which leaves a margin of only 33 observations. The study therefore fixes exclusion rules in advance and reports any shortfall. At the available size, the smallest detectable β is 0.594, essentially the planned 0.60. For RQ2 only 32 monthly points exist, so the design is limited. The smallest detectable level change at 80% power is 0.319 pp, about 61% of the pre-break mean TD of 0.525%. The observed raw pre–post difference (0.525% to 0.320%) is smaller than this threshold, so RQ2 is reported as an estimate with a confidence interval rather than as a decisive test. Low power is not its only limit: the estimate also depends on the model specification, which Table 4 documents. RQ3 is satisfied comfortably (342 versus 194), and RQ4 yields 69 recovery episodes against a minimum of 30.

Data Description

The data are NPCI's public "Top 50 Member Performance" tables, available at https://www.npci.org.in/product/ecosystem-statistics/upi (NPCI, n.d.). These are real, public, aggregate statistics, not synthetic data and not from a competition platform. NPCI publishes one table per month; the portal offers downloads from January 2024, and the 32 monthly files used here run from January 2024 to August 2026. Because NPCI does not permit automated access, files were downloaded manually and combined with a documented script. The remitter-side tables are used; the beneficiary-side tables are reserved as a robustness extension.

The compiled panel has 1,598 rows, 63 raw member names, and 56 harmonized entities, of which 39 appear in all 32 months. Thirty months list 50 members and two list 49 because two listings were combined. Preparation steps were: parsing percentages stored as text or fractions; harmonizing names with one rule-based map; combining duplicate listings within a month by volume weighting; and verifying that no two months share identical content. Two source files were found to carry the wrong month label; one was an exact duplicate and was dropped, and the other had different data and was relabeled by its file name and flagged. Non-bank members (credit-card issuers and payment technology firms, 43 rows) are flagged and excluded from bank-level models. Seventy rows report TD of exactly 0.00%; these are replaced with 0.0025 pp only when logits are computed. Table 2 defines every field.

Several integrity checks protect the panel. Each month has exactly one source file and each file maps to exactly one month. No bank appears twice in a month, no two months have identical volume vectors (the largest share of identical volume values between any two months is 2%), and the percentage identity holds in every row. These checks matter because several raw

files carried mislabeled titles, and a duplicated month would artificially inflate persistence and bias RQ4. The README and audit files in the repository record every correction.

Table 2
Data Dictionary

Field	Type	Definition
month	Date	Reporting month (first day), January 2024–August 2026
rank	Integer	NPCI rank of the member by volume in that month
bank_raw / bank	Text	Member name as published / harmonized name
entity	Text	Bank entity after merger and rename mapping (assumption documented)
volume_mn	Numeric	Attempted remitter transactions, millions
approved_pct	Numeric	Share of attempts approved (%)
bd_pct	Numeric	Business decline share (%)
td_pct	Numeric	Technical decline share (%); outcome variable
debit_reversal_mn	Numeric	Debit reversals, millions
debit_reversal_success_pct	Numeric	Share of debit reversals completed successfully (%)
is_bank	Logical	False for credit-card issuers and non-bank payment firms
month_source, source_file	Text	Provenance flags for the month label and the file used
Derived: lnV, ΔlnV, NewPeak, Post, Peak, Spike	Numeric	Log volume, log growth, prior-maximum indicator, post-August-2025 indicator, March/festive indicator, spike indicator (Analytic Approach)

Note. Approved + BD + TD = 100% for every row (checked). Source: NPCI (n.d.).

One mapping is a judgment call. Among bank rows, a single entity combines two names: Andhra Pradesh Grameena Vikas Bank (12 months in 2024) and Telangana Grameena Bank (20 months from January 2025), which the study treats as one bank after a rename. That affects 32 of 1,555 bank-months (2.1%). All other name changes are spelling variants of one bank, such as differences in case or punctuation. Every mapping is listed in an entity crosswalk file in the repository (entity_crosswalk.csv), and the rename is marked as an assumption to be verified against an official source before the final report. To check that the assumption cannot drive results, Table 3 re-estimates an illustrative post-August-2025 model with the entity mapped, left as two entities, and dropped. The estimates move by less than 0.03 logit units in the no-trend

model and less than 0.05 in the trend model, far less than their standard errors (about 0.09 and 0.33–0.35).

Table 3
Sensitivity of an Illustrative Post-August-2025 Estimate to the Entity Mapping

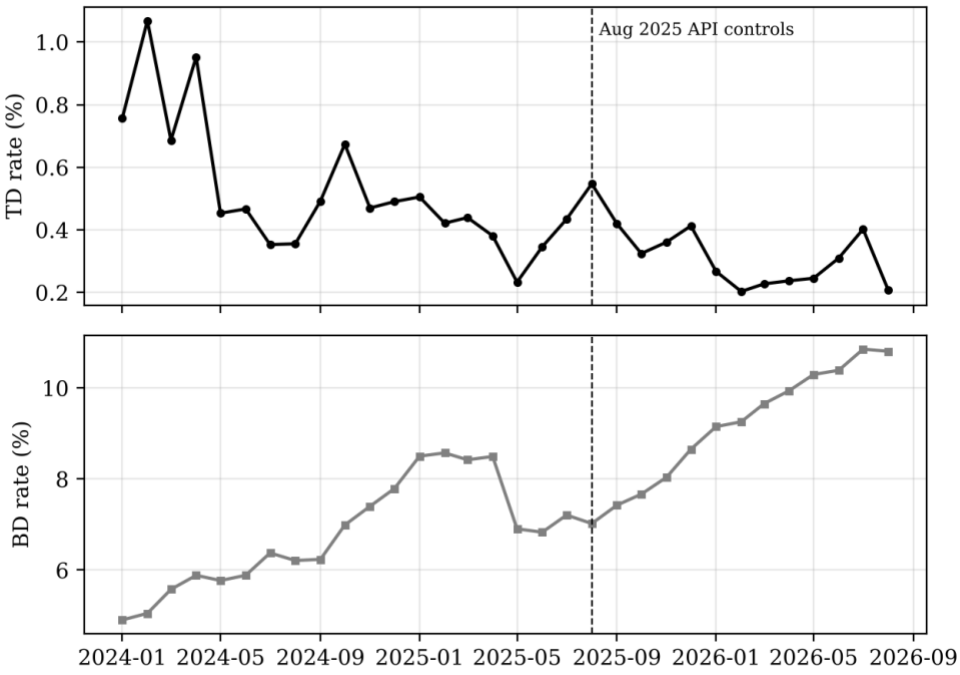
Treatment of renamed bank	Bank-months	Entities	Post coefficient, no trend (SE)	Post coefficient, with trend (SE)
Mapped as one entity (planned)	1,555	56	−0.470 (0.086)	0.577 (0.334)
Left as two entities	1,555	57	−0.484 (0.084)	0.583 (0.329)
Excluded	1,523	55	−0.499 (0.087)	0.624 (0.349)

Note. Fractional logit of TD share with entity fixed effects and entity-clustered standard errors (SE). Estimates test sensitivity to the mapping only and are not final results.

The full data, code, and a README are available in the project repository: https://github.com/karthick2613-git/ms_upi_capstone. Appendix A shows the folder tree and portal screenshots. Figure 1 plots the volume-weighted TD and BD rates across the window. TD fell from about 0.75% in January 2024 to between 0.20% and 0.54% since mid-2025, while BD rose from about 5% to about 10.8%. The decline in TD and rise in BD are descriptive facts that the research questions test.

Figure 1

Monthly Volume-Weighted Technical and Business Decline Rates, Top-50 Remitter Banks



Note. The dashed line marks August 2025, when NPCI's API controls took effect. Source: author's calculation from NPCI (n.d.).

Analytic Approach

The outcome is the technical decline share TD_{it} = td_pct / 100 for bank *i* in month *t*. Because TD is a proportion bounded by zero and one, with many values near zero, linear models on the raw share would give impossible fitted values. The study therefore uses fractional-response logit models (Papke & Wooldridge, 1996) for panel questions, interrupted time series for the system-level question (Bernal et al., 2017; Wagner et al., 2002), and survival and variance-components methods for persistence. Standard errors are clustered by bank, and because there are only about 56 clusters and a few treated groups, the wild cluster bootstrap is used for inference (Cameron et al., 2008). Hypotheses for each question appear under Research Questions 1–4; all tests are two-sided at $\alpha = .05$.

A spike is defined as a bank-month in which the robust z-score of TD exceeds 3, computed with the bank's own median and median absolute deviation, and TD is at least 1.0%. This rule flags 104 spike-months in 37 banks and yields 69 distinct episodes (runs of consecutive spike months). The threshold is a provisional choice, and alternatives ($z > 2.5$ and TD $\geq 0.75\%$) are used as sensitivity checks.

Causal-Inference Safeguards

The data are observational, so each question is framed as an association whose causal reading depends on stated assumptions. Three safeguards are applied. First, a placebo outcome: BD should not respond to the API controls, so a shift in BD would signal a general break rather than an effect on TD. Second, placebo break dates at every month with at least six months on each side (Figure 2) show whether the August 2025 estimate stands out. Third, bank and month fixed effects absorb stable bank differences and common shocks. The main threat to RQ2 is that the April 2025 outage, NPCI's other changes, and ordinary seasonality overlap in time, and with one system and no untreated group the design cannot separate them fully. RQ2 is therefore described as an evaluation of timing, not as proof of effect.

Diagnostics and Robustness

Each model carries checks aimed at its main weakness. Panel models: leave-one-bank-out estimates, winsorized TD, alternative floors for zero TD (0.001 and 0.01 pp), and exclusion of the three highest-volume banks. Time-series models: Ljung–Box and Breusch–Godfrey residual tests, alternative Newey–West lags, and a version excluding the April 2025 outage month. Survival models: alternative spike thresholds and recovery defined as two consecutive non-spike months. Results that do not survive these checks are described as fragile.

Feasibility, Measurability, and Data Availability

All variables are measurable: volume, approved share, BD share, and TD share are reported by NPCI for every listed member in every month, and every derived variable (log growth, prior-maximum indicator, calendar and intervention indicators, spike indicator) is computed from them. The data already exist in the compiled panel, so no collection risk remains. The analysis uses standard open-source libraries (Python and R) on a panel of fewer than 2,000 rows, which runs in minutes. The work fits within the 10-week capstone window: weeks 1–2 for data freeze and exploratory checks, weeks 3–5 for RQ1–RQ4 estimation, weeks 6–7 for robustness and the ML benchmark, weeks 8–9 for the report and repository, and week 10 for review. Before the final report, the beneficiary-side tables (and September 2026, if published) are added and the bank-rename mapping is verified against an official source. The data are public aggregate statistics from NPCI, were obtained from its official portal, and contain no personal information, so ethical sourcing is confirmed.

Threats to Validity, Ethics, and Limitations

Data are aggregated by bank and month, so findings describe banks, not individual users or transactions (ecological inference). The top-50 list changes over time, so the panel is unbalanced (39 of 56 entities are present in all months), and entering or leaving the list may relate to volume. The one bank-rename mapping is an assumption; it is documented in the repository and Table 3 shows it does not change the illustrative estimate. The sample margin for RQ1 is thin (1,555 against 1,522), and the RQ2 series has only 32 points, all banks were exposed on the same date, and its estimate depends on the model specification (Table 4). NPCI defines TD, and a change in classification practice would look like a real change; this cannot be ruled out from public data. The study uses only public, aggregate statistics with no personal data, so no

human-subject risk arises. Data are used with attribution to NPCI, NPCI's restriction on automated access is respected through manual download, and the repository records the source and date of every file.

Solution to RQ1–4

RQ1: Volume Shifts and TD

The model is a quasi-binomial fractional logit with bank and month fixed effects:

$$E[TD_{it} \mid x_{it}] = \Lambda(\alpha_i + \gamma_t + \beta_1 \ln V_{it} + \beta_2 \Delta \ln V_{it} + \beta_3 \text{NewPeak}_{it} + \beta_4 TD_{i,t-1})$$

where $\Lambda(u) = 1 / (1 + e^{-u})$, α_i are bank effects and γ_t are month effects. It is estimated by quasi-maximum likelihood with bank-clustered standard errors and wild cluster bootstrap *p*-values. Effects are reported as average marginal effects in percentage points and as odds ratios. Because lagged TD with fixed effects can bias short panels, the model is also run without the lag. Heterogeneity by bank group (public, private, small finance, and payments banks, coded by hand) is tested with interactions. An early fixed-effects check of the main volume slope returned a negative coefficient (−0.457, standard error [*SE*] = 0.212), which is why hypotheses are two-sided; it is a diagnostic and is not treated as a finding.

Expected output. Average marginal effects of a 10-percent volume jump and of a new peak on TD in percentage points, with bootstrap 95% confidence intervals, and a ranked list of banks whose TD is most sensitive to growth. A null or negative result is informative: it would suggest that capacity is not the binding constraint for those banks.

RQ2: Timing of the API Controls

The system-level series is monthly volume-weighted TD for banks. TD was falling for most of 2024, so the choice of baseline matters, and a straight line extrapolated from a falling, bounded series can overshoot. The study therefore estimates the August 2025 level change under

five specifications and not one: (A) a step with no trend; (B) a linear trend plus a step, the planned model shown below; (C) a step using only the 12 months before and after; (D) a step on log TD; and (E) a linear trend plus a step on log TD.

$$TD_t = \beta_0 + \beta_1 t + \beta_2 \, Post_t + \varepsilon_t$$

Here $Post_t = 1$ from August 2025 (19 pre-break and 13 post-break months). Standard errors are Newey–West (Newey & West, 1987) with lag 2, and intervals use the *t* distribution because the series is short. The same models are fitted to BD as a placebo outcome.

Small samples make asymptotic tests fragile, so each specification is also applied at every candidate break month that leaves at least six months on each side (21 candidates), and the rank of the August 2025 estimate among them is recorded. This is an in-time placebo test in the spirit of Abadie et al. (2010). Both the announcement (May 21, 2025, so June is the first full month) and enforcement (August 1) are examined, because banks and apps had notice before the rules took effect (Khanduri & Mittal, 2025).

The decision rule is fixed in advance. The study reports an August-specific association only if (i) the sign and 95% interval agree across all five specifications and (ii) the August estimate ranks in the top two of the 21 placebo breaks. If either condition fails, RQ2 is reported as inconclusive on attribution, which still tells NPCI what these public data cannot show.

Table 4 and Figure 2 report a feasibility check on the current data. It shaped the design and is not a final result. Without a trend (A, D), TD is lower after August 2025, but the August step ranks 20th and 19th of 21 breaks, because earlier breaks give larger drops while TD was already falling. With a trend (B, E), the sign reverses, since TD sits above the extrapolated pre-trend, and June and July 2025 give estimates as large as August. The 12-month window (C) shows a small decline of 0.12 pp. On these data the second condition of the decision rule fails, so

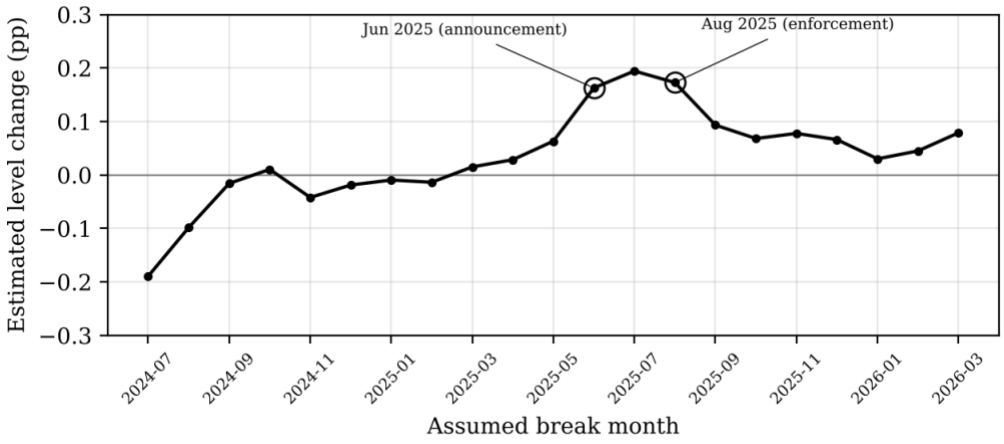
RQ2 is expected to be inconclusive on attribution: TD improved and flattened around mid-2025, but the data do not separate the August 1 rules from that wider pattern.

Table 4
Preliminary Specification Check for the August 2025 Level Change in TD

Specification	Estimate	95% CI	Rank of August (of 21)
A. Step only (no trend)	−0.205 pp	−0.361 to −0.049	20
B. Linear trend + step (planned)	+0.173 pp	0.045 to 0.301	3
C. 12 months before vs. after, step	−0.116 pp	−0.222 to −0.011	n/a
D. Log TD, step only	−37.7%	−54.9% to −13.7%	19
E. Log TD, linear trend + step	+30.2%	2.8% to 64.8%	3

Note. Volume-weighted monthly TD, 32 months; Newey–West standard errors (lag 2) with *t* intervals; pp = percentage points. Rank is by absolute size among the 21 candidate break months (1 = largest); a rank near 1 would mark August as unusual.

Figure 2
Estimated Level Change in TD by Assumed Break Month (Linear Trend plus Step)



Note. Each point re-fits the planned model with the break at that month. Circles mark June 2025 (announcement) and August 2025 (enforcement).

Two designs use the bank-level panel without leaning on its small standard errors for a shock that hit every bank on the same date. A dose-response check asks whether the post-period change differs with a bank's pre-period size, with month fixed effects absorbing common shocks; the preliminary coefficient is −0.099 per standard deviation of log volume (*SE* = 0.087, 49 banks), not distinguishable from zero, and it is treated as exploratory. BD serves as a placebo

outcome: with a trend it shows no level shift (−0.93 pp, 95% CI −2.70 to 0.84), but without a trend it rises by 2.4 pp because BD has trended up since 2024, so it is a valid placebo only in the trend-adjusted form. As a further design, the beneficiary-side tables will be added as a comparison series for a difference-in-differences contrast with remitter-side TD. The controls target calls made by apps to remitter-side services, so remitter-side TD should respond more; that is an assumption and will be stated as one.

Expected output. A table of the five specifications with 95% intervals, the placebo-break distribution and the August rank, the BD placebo, and a plain statement of whether the decision rule is met. A wide or inconsistent result is reported as inconclusive, not as evidence of no effect; the smallest detectable shift is 0.319 pp.

RQ3: Calendar Peaks

An event-study fractional logit with bank fixed effects includes indicators for event months around each March fiscal year-end (March 2024, 2025, 2026; one month before to one month after) and for the festive months (October–November):

$$E[TD_{it}] = \Lambda(\alpha_i + \delta\ Peak_t + \theta\ (Peak_t \times \ln\bar{V}_i) + controls)$$

where $\ln\bar{V}_i$ is the bank's prior-year mean log volume, a measure of exposure; δ tests $H0_{3a}$ and θ tests $H0_{3b}$. The 2025 outages fall near the March 2025 event, so results are reported with and without that year.

Expected output. Event-study coefficients by month relative to each March year-end and for the festive months, a peak-by-exposure interaction, and a table of which peak months show elevated TD for which kinds of bank. If only March 2025 shows an effect, the result will be attributed to the outage period and not to the calendar.

RQ4: Persistence and Heterogeneity

Three tools are used. A random-intercept model, $\ln(TD/(1-TD))_{it} = \mu + u_i + e_{it}$, gives the ICC = $\sigma^2_u / (\sigma^2_u + \sigma^2_e)$. A panel AR(1) coefficient ρ gives a half-life of $\ln(0.5) / \ln(\rho)$ months. Kaplan–Meier curves (Kaplan & Meier, 1958) and a Cox model (Cox, 1972) describe how long spike episodes last, where the event is recovery to a non-spike month. In the preliminary computation on the full panel, the ICC is 0.557 ($\sigma^2_u = 1.591$, $\sigma^2_e = 1.264$) and $\rho = 0.318$, giving a half-life of 0.605 months. Of 69 episodes, all recovered, with a median duration of one month and a maximum of four. These figures size the study; they are not final estimates.

Expected output. The ICC and half-life with confidence intervals, Kaplan–Meier curves of spike duration by bank group, and Cox hazard ratios for recovery. A high ICC (the preliminary value is 0.557) would mean that a bank's own history explains over half of the variation in TD, which would shift attention from system-wide capacity toward individual bank remediation.

Machine Learning Benchmark

Inside RQ1 and RQ4, without changing them, a gradient-boosted tree model (XGBoost; Chen & Guestrin, 2016) is benchmarked on two tasks: regression of next-month TD, and classification of next-month spikes. Eight lagged or calendar features are used: lagged TD, lagged BD, $\ln V$, $\Delta \ln V$, NewPeak, sine and cosine of the month, and bank group. Only information available at month t predicts month t+1. Validation is rolling-origin, in which the training window expands and the model forecasts the following month (Bergmeir & Benítez, 2012), with January–August 2026 held out. The comparators are a naive last-month rule and the RQ1 fractional logit. Predictive results show whether the statistical models miss structure; they say nothing about causes. Because the hold-out contains only five spikes, precision–recall

measures are noisy and are reported alongside the rolling-origin results (Saito & Rehmsmeier, 2015).

Evaluation Metrics

Table 5 gives each metric, its formula, and the value computed on the real data so far. Worked values come from the naive baseline on the January–August 2026 hold-out (n = 383 bank-months) and from the preliminary survival and variance calculations.

Table 5
Evaluation Metrics, Formulas, and Worked Values

Metric	Formula	Worked value (current data)
RMSE	$\sqrt{(}\ (1/n)\ \Sigma\ (y_i - \hat{y}_i)^2\ )$	Naive ($TD_t = TD_{t-1}$): 0.929 pp, the benchmark to beat
MAE	$(1/n)\ \Sigma\ \lvert y_i - \hat{y}_i \rvert$	Naive: 0.434 pp
Precision	TP / (TP + FP)	Naive spike rule: 2 / (2 + 3) = 0.40
Recall	TP / (TP + FN)	Naive spike rule: 2 / (2 + 3) = 0.40
PR-AUC	$AP = \Sigma_n\ (R_n - R_{n-1})\ P_n$	Computed for XGBoost and the logit; the 5 hold-out spikes make it noisy
Kaplan–Meier	$S(t) = \prod\ (1 - d_j/n_j)$	$S(1) = 1 - 47/69 = 0.319$; $S(2) = 0.319 \times (1 - 13/22) = 0.130$
ICC	$\sigma^2_u\ /\ (\sigma^2_u + \sigma^2_e)$	1.591 / (1.591 + 1.264) = 0.557
AR(1) half-life	$\ln(0.5)\ /\ \ln(\rho)$	ln(0.5) / ln(0.318) = 0.605 months
ITS level change	$\beta_2 \pm 1.96 \times SE(\beta_2)$	Design precision: 1.96 × 0.114 = ±0.223 pp

Note. RMSE = root mean squared error; MAE = mean absolute error; PR-AUC = area under the precision–recall curve; ICC = intraclass correlation; ITS = interrupted time series; AR(1) = first-order autoregressive; TP, FP, FN = true positives, false positives, false negatives; d_j = events and n_j = at risk at time j; R = recall; P = precision.

Model comparison for RQ1 and RQ3 uses quasi-likelihood-based information measures and out-of-sample RMSE. Confidence intervals come from bank-clustered bootstrap draws, and multiplicity is handled by treating the two RQ1 hypotheses and the RQ3 peak-by-exposure term as a small family with Holm adjustment.

Recommendation and Application

The primary users are NPCI's product and operations teams, who set API rules and monitor member performance, and remitter banks' payments infrastructure teams, who decide capacity. A secondary audience is the Reserve Bank of India's payment-systems oversight function. If RQ1 and RQ3 show that TD rises in particular banks after abrupt growth or in named peak months, capacity can be targeted at those banks ahead of March and the festive season. RQ2 can inform but cannot by itself justify keeping or relaxing the API controls: if its decision rule is met, the result is evidence of timing consistent with an effect; if not, as the preliminary check suggests, NPCI should evaluate the controls with finer data, such as daily or transaction-level TD around the rule dates. RQ4 informs how fast to expect recovery: if spikes last about a month, as the preliminary episodes suggest, monitoring should trigger action within one reporting cycle. The machine learning benchmark indicates whether a lightweight early-warning flag based on lagged public data is feasible. Public monthly data are coarse, and recommendations are written as guidance for further internal analysis with transaction-level data.

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Appendix A

Repository Structure and Data Screenshots

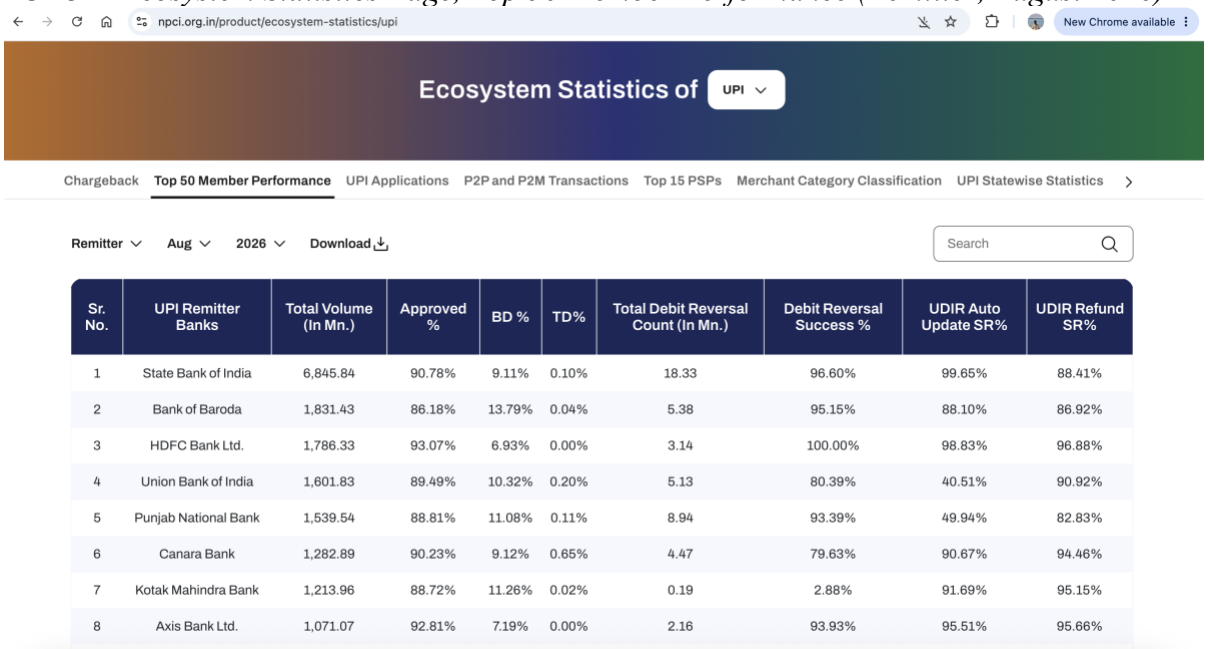
Repository: https://github.com/karthick2613-git/ms_upi_capstone

Folder tree (planned):

```
capstone/
  data/raw/          32 monthly NPCI remitter files
  data/processed/    upi_remitter_top50_panel.csv
                     file_log.csv, month_audit.csv
  code/              combine_remitter.py, sample_size.py,
                     analysis scripts
  outputs/           tables and figures
  README_data_notes.md
```

Figure A1

NPCI UPI Ecosystem Statistics Page, Top 50 Member Performance (Remitter, August 2026)



Sr. No.	UPI Remitter Banks	Total Volume (In Mn.)	Approved %	BD %	TD%	Total Debit Reversal Count (In Mn.)	Debit Reversal Success %	UDIR Auto Update SR%	UDIR Refund SR%
1	State Bank of India	6,845.84	90.78%	9.11%	0.10%	18.33	96.60%	99.65%	88.41%
2	Bank of Baroda	1,831.43	86.18%	13.79%	0.04%	5.38	95.15%	88.10%	86.92%
3	HDFC Bank Ltd.	1,786.33	93.07%	6.93%	0.00%	3.14	100.00%	98.83%	96.88%
4	Union Bank of India	1,601.83	89.49%	10.32%	0.20%	5.13	80.39%	40.51%	90.92%
5	Punjab National Bank	1,539.54	88.81%	11.08%	0.11%	8.94	93.39%	49.94%	82.83%
6	Canara Bank	1,282.89	90.23%	9.12%	0.65%	4.47	79.63%	90.67%	94.46%
7	Kotak Mahindra Bank	1,213.96	88.72%	11.26%	0.02%	0.19	2.88%	91.69%	95.15%
8	Axis Bank Ltd.	1,071.07	92.81%	7.19%	0.00%	2.16	93.93%	95.51%	95.66%

Figure A2

First Rows of the Combined Remitter Panel (upi_remitter_top50_panel.csv)

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
month	file_month	rank	bank_raw	bank	volume_mn	approved_pc	bd_pct	td_pct	debit_reversa	debit_reversa	source_file	month_sourc	n_member_r	entity	is_bank	pct_sum
2024-01	2024-01	1	State Bank O	State Bank of	3128.97	95.62	4.19	0.19	11.58	96.85	b637e410-Ec	sheet title	1	State Bank of	TRUE	100
2024-01	2024-01	2	HDFC BANK	HDFC Bank	1040.25	96.49	3.51	0	3.34	98.84	b637e410-Ec	sheet title	1	HDFC Bank	TRUE	100
2024-01	2024-01	3	Bank of Baro	Bank of Baro	781.72	95.28	4.68	0.04	2.71	98.54	b637e410-Ec	sheet title	1	Bank of Baro	TRUE	100
2024-01	2024-01	4	Union Bank o	Union Bank o	753.66	94.51	4.34	1.15	7.16	46.76	b637e410-Ec	sheet title	1	Union Bank o	TRUE	100
2024-01	2024-01	5	Punjab Natio	Punjab Natio	618.95	94.78	4.95	0.27	2.61	88.58	b637e410-Ec	sheet title	1	Punjab Natio	TRUE	100
2024-01	2024-01	6	Kotak Mahinc	Kotak Mahinc	600.83	93.64	5.19	1.17	2.08	89.49	b637e410-Ec	sheet title	1	Kotak Mahinc	TRUE	100
2024-01	2024-01	7	Canara Bank	Canara Bank	572.16	94.18	5.38	0.44	3.02	85.98	b637e410-Ec	sheet title	1	Canara Bank	TRUE	100
2024-01	2024-01	8	Axis Bank Ltd	Axis Bank	559.49	96.11	3.58	0.3	2.66	60.21	b637e410-Ec	sheet title	1	Axis Bank	TRUE	99.99
2024-01	2024-01	9	ICICI Bank	ICICI Bank	547.55	96.59	3.32	0.08	1.64	95.72	b637e410-Ec	sheet title	1	ICICI Bank	TRUE	99.99
2024-01	2024-01	10	Paytm Payme	Paytm Payme	410.06	96.3	3.7	0.01	0.87	98.11	b637e410-Ec	sheet title	1	Paytm Payme	TRUE	100.01
2024-01	2024-01	11	Bank of India	Bank of India	389.29	95.33	4.59	0.08	1.48	91.49	b637e410-Ec	sheet title	1	Bank of India	TRUE	100
2024-01	2024-01	12	Indian Bank	Indian Bank	348.89	95.15	4.47	0.37	1.51	84.37	b637e410-Ec	sheet title	1	Indian Bank	TRUE	99.99
2024-01	2024-01	13	Airtel Payme	Airtel Payme	337.05	89.95	8.98	1.07	1.93	69.9	b637e410-Ec	sheet title	1	Airtel Payme	TRUE	100
2024-01	2024-01	14	India Post Pa	India Post Pa	265.13	82.66	12.2	5.14	1.67	54.81	b637e410-Ec	sheet title	1	India Post Pa	TRUE	100
2024-01	2024-01	15	Federal Bank	Federal Bank	185.9	95.53	4.28	0.19	0.65	93.72	b637e410-Ec	sheet title	1	Federal Bank	TRUE	100
2024-01	2024-01	16	Central Bank	Central Bank	178.41	85.44	9.38	5.18	4.29	72.15	b637e410-Ec	sheet title	1	Central Bank	TRUE	100
2024-01	2024-01	17	Indian Overs	Indian Overs	165.09	94.28	4.41	1.31	0.99	90.36	b637e410-Ec	sheet title	1	Indian Overs	TRUE	100
2024-01	2024-01	18	UCO Bank	UCO Bank	152.41	88.37	7.47	4.16	2.33	40.45	b637e410-Ec	sheet title	1	UCO Bank	TRUE	100
2024-01	2024-01	19	IDBI Bank Lin	IDBI Bank	141.57	92.73	4.28	2.98	1.31	81.4	b637e410-Ec	sheet title	1	IDBI Bank	TRUE	99.99
2024-01	2024-01	20	Bank of Maha	Bank of Maha	136.29	94.42	4.61	0.97	0.61	76.33	b637e410-Ec	sheet title	1	Bank of Maha	TRUE	100
2024-01	2024-01	21	INDUSIND BA	IndusInd Ban	125.63	93.06	5.97	0.97	0.68	98.66	b637e410-Ec	sheet title	1	IndusInd Ban	TRUE	100
2024-01	2024-01	22	Fino Paymen	Fino Paymen	115.14	89.66	9.41	0.93	0.63	67.81	b637e410-Ec	sheet title	1	Fino Paymen	TRUE	100
2024-01	2024-01	23	Karnataka Ba	Karnataka Ba	97.36	91.93	7.49	0.59	0.42	77.55	b637e410-Ec	sheet title	1	Karnataka Ba	TRUE	100.01
2024-01	2024-01	24	Yes Bank Ltd	YES Bank	93.47	94.92	4.98	0.1	0.25	93	b637e410-Ec	sheet title	1	YES Bank	TRUE	100
2024-01	2024-01	25	IDFC FIRST B	IDFC FIRST B	73.92	94.2	5.77	0.03	0.03	0	b637e410-Ec	sheet title	1	IDFC FIRST B	TRUE	100
2024-01	2024-01	26	Karur Vysya B	Karur Vysya B	59.56	94.12	5.54	0.34	0.24	92.86	b637e410-Ec	sheet title	1	Karur Vysya B	TRUE	100
2024-01	2024-01	27	Bandhan Bar	Bandhan Bar	49.22	87.51	6.16	6.32	0.85	68.2	b637e410-Ec	sheet title	1	Bandhan Bar	TRUE	99.99
2024-01	2024-01	28	South Indian	South Indian	48.9	96.35	3.55	0.11	0.14	95.97	b637e410-Ec	sheet title	1	South Indian	TRUE	100.01
2024-01	2024-01	29	Jammu and K	Jammu & Kas	44.46	91.55	7.21	1.24	0.31	70.87	b637e410-Ec	sheet title	1	Jammu & Kas	TRUE	100